

Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Recent advances in artificial intelligence (AI) have enabled widespread use of AI assistants for various cognitive tasks, ranging from everyday tasks to scientific discovery and creative writing. While AI assistance can improve the performance of each individual, it can weaken their subsequent performance without AI and further limit the diversity of interests and opinions. These risks raise the question of whether human collective intelligence can persist as our society becomes increasingly reliant on AI assistants. To address this question, we develop an evolutionary model of collective intelligence, where each individual participate in a prediction task, consult AI, and imitate each other's prediction strategy. We find that "Chatbot" AI, which depends on users' interests, can prevent the emergence of collective intelligence by either limiting diversity or propagating systematic bias through overreliance. Directly incentivizing or penalizing AI use generally worsens these outcomes. Instead, a balanced incentive structure that preserve minority expertise and reward contributions to collective outcome can promote the emergence of collective intelligence. However, the inherent ability of the population's to make prediction in the absence of AI remains suboptimal even under this balanced incentive structure, raising concerns for continued AI reliance.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007; Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004; Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and computer science (Werfel, 2026). Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes (Cui and Yasseri, 2024).

The recent emergence of generative AI and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi and Hauser, 2024). In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the potential long-term impact of AI assistants on human society, especially CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise (Vaccaro et al., 2024) and opinion aggregation (Burton et al., 2024; Tessler et al., 2024). Conversely, empirical evidence indicates that the use of AI can limit the diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung, 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

Our ability to predict the long-term impact of AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI use. On one hand, many industrial sectors are pushing towards increased AI use, where the number of “tokens” used is often perceived as a proxy for productivity (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such discrepancy raises a question about whether directly incentivizing or penalizing AI use will benefit or harm human CI. More broadly, previous theoretical work identified incentive structures that promote CI through social learning (Wang et al., 2025), but it remains unclear whether these structures remain effective in the presence of AI assistants. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of AI assistants. Such incentive structure should seek to retain the benefits of AI and prevent potential risk, such as preserving

the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, in which each individual participates in a prediction task based on partial observations of underlying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012). In particular, we allow each individual to independently consult with AI assistants and revise their predictions accordingly. Individual interests, beliefs, and reliance on AI then evolve through imitation of peers with higher payoffs. We show that incentive structures that promote CI without AI can become ineffective in its presence. We examine the how AI use affect interest diversity and collective bias, and leverage this understanding to propose incentive structures that can maintain CI in the presence of AI.

Mathematical model

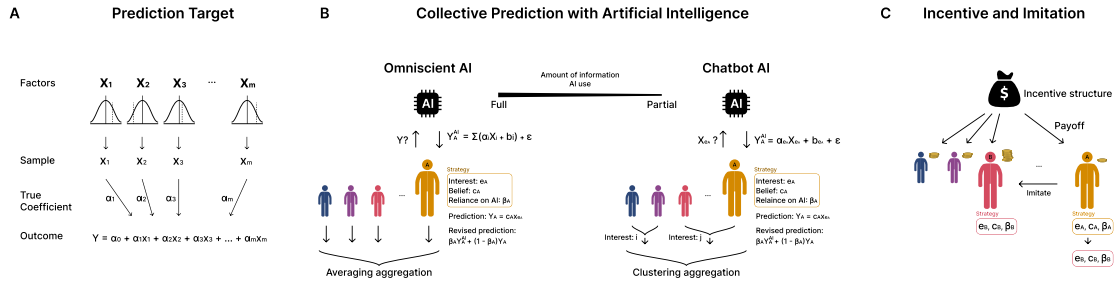


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

We begin by extending previously established models of CI, in which a group of players participates in a prediction task (Fig. 1): each player makes an individual prediction based on partial observations of the underlying causal factors, and the group forms a collective prediction by aggregating these individual predictions (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction target, Y , is modeled as a linear combination of M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution (Fig. 1A). Then, each individual A observes a single factor of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A =$

90 $c_A x_{e_A}$ (Fig. 1B). We assume each individual can then independently consult their
 91 AI assistants and revise their prediction based on their reliance on AI, β_A : $Y_A^{re} =$
 92 $\beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Fig. 1B). The revised predictions are then aggregated using two
 93 alternative aggregation rules, averaging and clustering, to form collective prediction.

94 To model AI, we assume that AI has already been trained on similar tasks and
 95 is able to make predictions based on partial or full observations of causal factors
 96 (Fig. 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based
 97 on its internal parameters, which can include factor-specific bias (b_i) and stochastic
 98 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 99 amount of information AI use when making predictions. In one extreme, the AI
 100 can observe all causal factors and use all information to make predictions, regardless
 101 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 102 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 103 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 104 and ‘Chatbot AI’, respectively.

105 We assume that appropriate aggregation rule for collective prediction depends
 106 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 107 single factor. Thus, we use clustering aggregation, under which revised predictions
 108 are first averaged among individuals who are interested in the same factor, and
 109 then summed across factors. When Omniscient AI is used, however, each revised
 110 prediction contains information about all causal factors, in which case clustering
 111 these predictions by interest would be inappropriate. Instead, we use averaging
 112 aggregation, under which revised individual predictions are averaged directly to form
 113 the collective prediction.

114 Each generation consists of multiple round of predictions, and each player is re-
 115 warded a payoff according to the incentive structure (Fig. 1C). For simplicity, we
 116 assume that a sufficiently large number of prediction rounds occurs within each gen-
 117 eration, allowing us to use the expected payoff value rather than computing the
 118 payoffs for every round. We consider two incentive structures from previous work,
 119 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 120 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 121 gives higher payoff to individuals who can push the collective prediction toward the
 122 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{re}(Y - \hat{Y}^{re})$. In
 123 contrast, the niche-expert structure gives higher payoff to individuals who have both
 124 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{re} - Y)^2$. Here, ρ_{e_A}
 125 denotes the proportion of players interested in the same factor, thereby allowing the
 126 niche-expert structure to preserve interest diversity. We note collective intelligence
 127 fails to emerge under the combination of averaging aggregation with niche-expert
 128 structure in the absence of AI (Wang et al., 2025).

129 Through time, the group evolves by allowing each individual to imitate peers
 130 who received higher payoffs (Fig. 1C). Specifically, after each generation, two indi-
 131 viduals are randomly sampled, and one imitates all three strategies (the factor of
 132 interest, factor belief, and reliance on AI) of the other with a probability determined

by the difference in their expected payoffs (Materials and Methods). Individual predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation rules. Collective accuracy is defined as the reduction in prediction error achieved by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025): $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective accuracy reaches a near perfect level. Full derivations of the collective prediction, expected payoffs, and collective accuracy under the two AI models are provided in Supplementary Text S1 and S2.

This model now allows us to explore the impact of AI on CI depending on the type of AI and the incentive structure. Here, we begin by comparing the emergence of CI with and without AI and subsequently investigate how different incentive structures affect the emergence of CI in the presence of AI.

AI can either promote or prevent the emergence of CI

To understand the impact of AI, we first compare collective dynamics with and without AI. First, consistent with Wang et al. (2025), simple imitation rules and incentive structures allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One exception is the use of simple averaging with niche-expert incentive structures, under which the collective accuracy of the population remains near zero. This is because enhancing individual accuracy causes individuals to predict close to the true coefficient of their interest, but simply averaging these predictions does not necessarily produce an accurate estimate of the full prediction target (Fig. S1).

We find that AI use can have heterogeneous outcomes depending on its type. For example, the use of Omniscient AI can either preserve CI under the feedback incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-expert incentive structure (Fig. 2B). These results are robust across different levels of AI accuracy (Fig. S2) and can also be characterized analytically in the large-population limit (Supplementary Text S3.1).

In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive structure, the use of Chatbot AI leads to a loss of diversity in the population's interest, making it impossible to account for effect of some causal factors (Fig. S2, S3). This causes an increase in collective variance and stochastic decrease in collective accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-expert incentive structure, the population becomes overly reliant on AI, causing the collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly, we find that the niche-expert structure can prevent the use of AI under specific

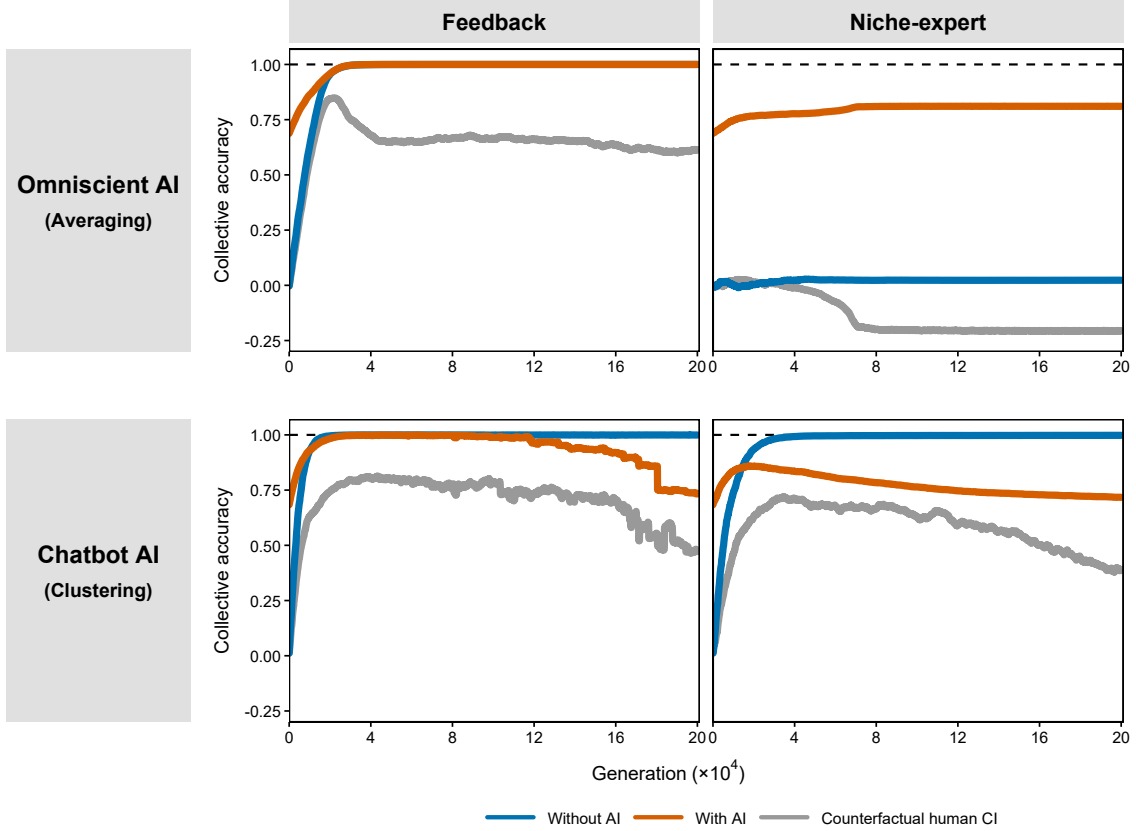


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

assumptions about the relationship between the true coefficient and AI bias (Fig. S2). In this case, the population reduces the reliance on AI assistance, allowing the CI to emerge eventually. However, this phenomenon can substantially delay the emergence of CI (Fig. S5).

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at any given time, we quantify counterfactual collective accuracy based on unrevised predictions. Across all scenarios, we find that

the counterfactual collective accuracy is lower than the collective accuracy of the corresponding population (Fig. 2).

In summary, an Omniscient AI can promote CI, but AI operating with limited information may prevent the emergence of CI. Moreover, reliance on AI can limit inherent ability of the population to make accurate predictions on their own, raising concerns about future AI use. These findings further underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

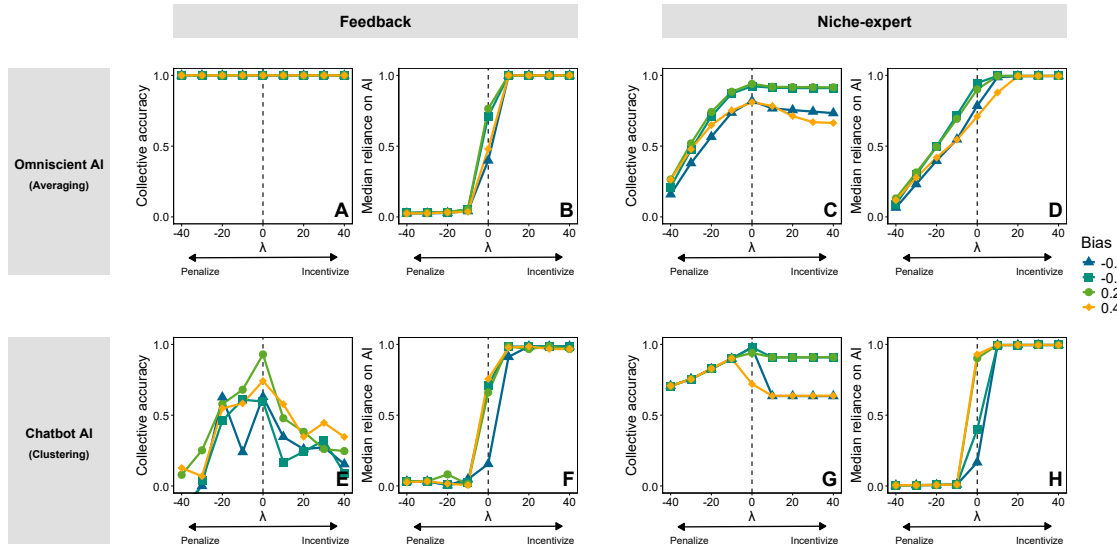


Figure 3: Collective accuracy under the influence of direct intervention on AI use. Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (Omniscient AI, A–D and Chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

Given the strong influence of AI on collective outcomes, a straightforward inter-

vention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A represent individual reliance on AI (Supplementary Text S1.3 and S2.3).

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI use causes individuals to imitate others based on their AI use rather than their predictive accuracy. Thus, stronger interventions can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G). Overall, these results indicate that directly rewarding or penalizing AI use, independently of predictive performance, may exacerbate rather than mitigate the risk of AI on CI.

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Here, we propose a “balanced” incentive structure, modeled as the average of the feedback and niche expert incentives (Supplementary Text S2.3).

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to the loss of interest diversity but can prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However the niche-expert structure cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which in turn homogenize the predictions.

Therefore, when the two incentive structures are combined, the population achieves CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Under the balanced structure, individuals remain more evenly distributed across causal factors, and their beliefs lie closer to the true coefficients than under either incentive structure alone (Fig. 4C–E). The balanced structure also produces higher counterfactual collective accuracy than the feedback or niche-expert structure (Fig. 4B, S6). These patterns are robust to how the two incentive structures are weighted

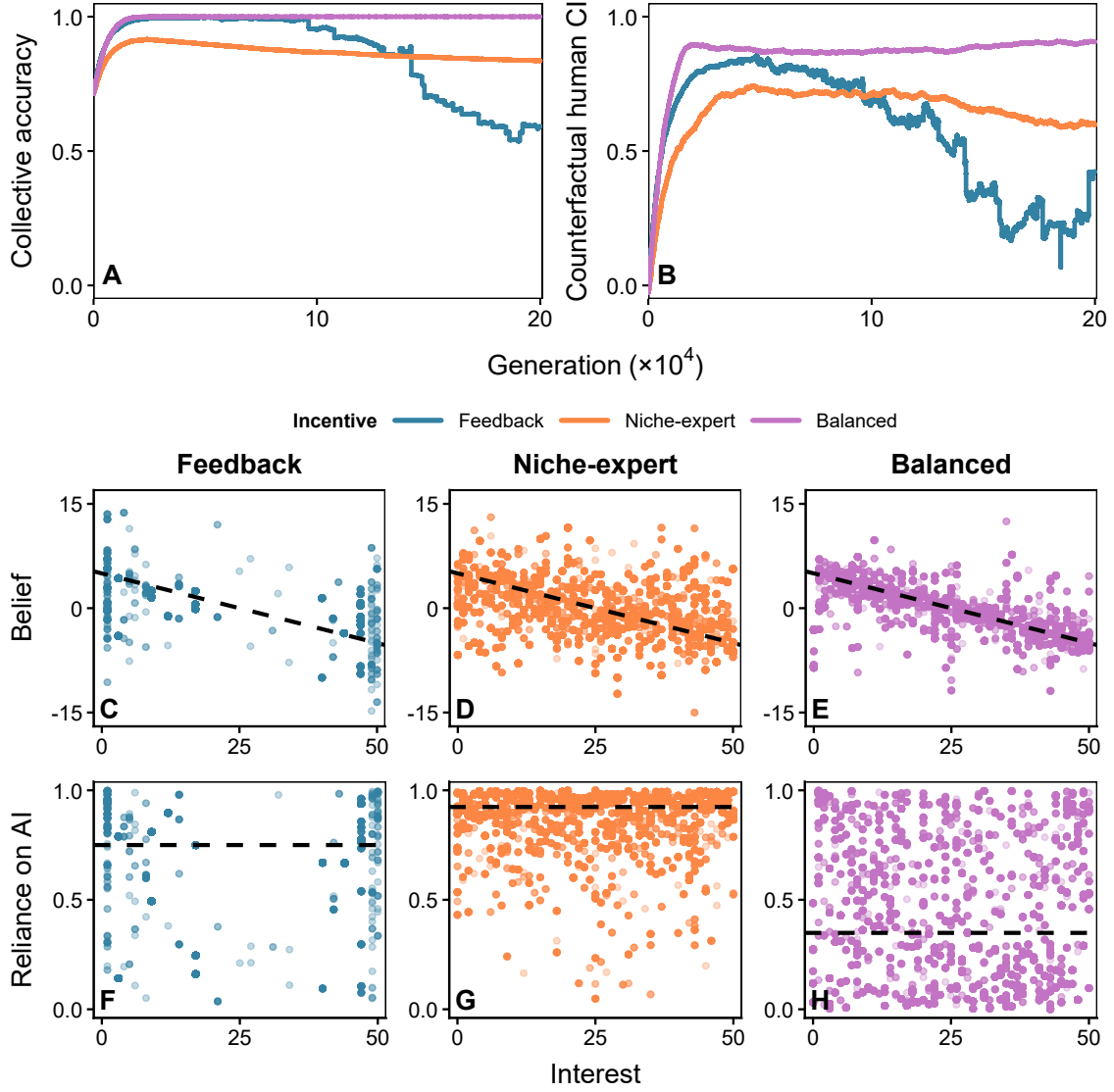


Figure 4: **Balanced structure can promote collective intelligence.** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

when combined (Fig. S7), and are driven by the emergent property of the balanced structure, under which the population increases its reliance on AI as AI becomes

231 more accurate (Fig S8).

232 Discussion

233 The ongoing race towards the development of “better” AI continues to pose ques-
234 tions about the societal benefits and risks of AI use (Bender et al., 2021; Burton
235 et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolu-
236 tionary dynamics to investigate how AI use and incentive structures jointly affect
237 the emergence of CI, which is defined as the population’s ability to make accurate
238 predictions. We find that an “Omniscient” AI model, which incorporates informa-
239 tion about all relevant causal factors, can improve collective accuracy, whereas a
240 “Chatbot” AI model, whose outcome depends on the user’s interest, can impair it.
241 These contrasting outcomes arise because AI assistance affects not only individual
242 predictions, but also the population’s interest diversity, beliefs, and reliance on AI.
243 We also show that directly incentivizing or penalizing AI use, independent of indi-
244 vidual’s performances, can further limit the collective accuracy of the population.
245 Instead, a balanced incentive structure can maintain CI by promoting diversity and
246 limiting bias.

247 Our analysis reveals that AI can prevent the emergence of CI through two differ-
248 ent mechanisms. First, AI use can cause individuals to concentrate on a subset of the
249 factors that determine the prediction target, thereby reducing diversity in their inter-
250 ests. This result is broadly consistent with recent empirical studies showing that AI
251 assistance can improve individual performance but limit the diversity of outputs, in
252 domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi
253 and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit
254 inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023),
255 causing the prediction accuracy to decrease eventually. This mechanism can prevent
256 the emergence of CI even when the interest diversity is maintained because biases in
257 AI predictions can be amplified when individual predictions are aggregated. Thus,
258 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI use
259 need to be calibrated to prevent systematic AI biases from becoming collective biases
260 (Bućinca et al., 2021).

261 Our analysis of counterfactual collective accuracy also raises an important concern
262 for future AI use: while AI use may improve the apparent performance of individuals
263 and, in some cases, the population as a whole, it can simultaneously reduce the pop-
264 ulation’s ability to make accurate predictions without AI. These results echo findings
265 from a recent field experiment, which demonstrated that students, who seemingly
266 performed better in solving mathematics problems under AI assistance, performed
267 worse on their own than those who did not have access (Bastani et al., 2025). These
268 findings highlight the need to evaluate AI tools not only by the performance gains
269 they provide during use, but also by whether individuals and populations retain the
270 ability to perform without them (Melumad and Yun, 2025; Ke et al., 2026). Making

271 this distinction clear will be particularly important for education, where AI may
272 help students complete a task without helping them learn how to do it on their own
273 (Bastani et al., 2025; Melumad and Yun, 2025).

274 As generative AI and large language models are already generating substantial
275 benefits, many organizations, both industrial and academic, are increasingly seeking
276 to improve their collective performance by regulating AI use (Thorp, 2023; Bous-
277 quette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct forms,
278 including AI-use policies or token-based leaderboards, or may operate indirectly
279 through other incentives, such as productivity rewards or the social stigma asso-
280 ciated with AI use (Reif et al., 2025). Our model suggests that directly incentivizing
281 or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI
282 or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
283 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
284 venting groups from using AI even when it performs well in the relevant domain.
285 Our result instead shows that appropriate AI use and accurate CI can be promoted
286 by focusing on the traditional properties of groups that have been emphasized for
287 decades: maintaining diversity and preventing collective bias, rather than on AI use
288 itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

289 There are several limitations to our analyses. First, our model allows for social
290 learning among humans, but AI’s inherent ability remains fixed throughout the evo-
291 lutionary process. In reality, AI can also learn from human-generated data (Ouyang
292 et al., 2022). Such bidirectional learning could increase the similarity between hu-
293 man and AI errors and thereby reduce group performance (Glickman and Sharot,
294 2025). Second, we model AI errors using fixed, factor-specific additive biases and
295 independent random errors. This assumption allows random errors to be reduced
296 through aggregation, while systematic biases can still persist in the collective pre-
297 diction. In reality, AI errors may have more complex structures, including biased
298 estimates of factor effects and correlated errors across individuals, factors, or re-
299 peated interactions. Further empirical and theoretical work is therefore needed to
300 understand how these error structures affect the emergence of CI. Finally, following
301 established models of CI, we assume that the prediction target is a linear combination
302 of independent causal factors. This assumption does not hold for many real-world
303 tasks, where causal factors may be correlated or interact nonlinearly. Future studies
304 should therefore investigate how nonlinear relationships between causal factors and
305 prediction targets affect the interaction between AI assistance and the emergence of
306 CI.

307 The rapid adoption of AI assistants is changing how individuals learn, work, and
308 make decisions (Tessler et al., 2024; Glickman and Sharot, 2025; Sajadieh et al.,
309 2026; Bick et al., 2026; Ke et al., 2026). While the immediate benefits and risks
310 of these systems have been evaluated at the individual level (Noy and Zhang, 2023;
311 Bastani et al., 2025), our study raises a broader concern that their widespread AI use
312 may reshape collective intelligence of human society. Our results further underline
313 the importance of designing reward structures that preserve minority expertise and

314 prevent AI biases from persisting across the population (Mann and Helbing, 2017;
 315 Vicente and Matute, 2023; Glickman and Sharot, 2025; Wang et al., 2025). A de-
 316 tailed empirical understanding of how AI assistance reshapes diversity, expertise, and
 317 reliance at the population level will therefore be critical for predicting the dynamics
 318 of future of human collective intelligence.

319 Methods

320 Prediction target

321 Following previous models of CI, we model the prediction target as a linear combi-
 322 nation of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017;
 323 Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

324 where the coefficient α_i determines the magnitude and direction of the effect of
 325 factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an
 326 independent Gaussian random variable with mean zero and standard deviation σ_i .

327 Players

328 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
 329 prediction task. In the absence of AI assistance, the strategy of player A is char-
 330 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
 331 factor X_{e_A} and makes a prediction by multiplying the observed factor value by their
 332 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

333 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does
 334 not observe any causal factor and instead makes a constant prediction based solely
 335 on their belief. We refer to this as a stubborn strategy because the player’s predic-
 336 tion does not respond to the realized values of the causal factors. For notational
 337 convenience, we assume that $X_0 = 1$.

338 When the group is assisted by AI, each player’s strategy additionally includes
 339 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 340 pendently and revises their prediction by taking a weighted average of their own
 341 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

342 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI,
 343 whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

Second, when the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised

379 prediction therefore contains information about only one factor and can be treated
 380 as an estimate of that factor’s contribution to the target. We consequently use
 381 clustering aggregation.

382 Clustering aggregation is performed by grouping individuals according to their
 383 interests and calculating the mean prediction within each group. The collective
 384 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

385 where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents
 386 the number of players in that set.

387 When the group is not assisted by AI, collective predictions are computed from the
 388 unrevised individual predictions throughout the simulation. Both simple averaging
 389 and clustering can be used in the absence of AI. For fair comparison, we use the
 390 same aggregation rule as in the corresponding AI-assisted condition.

391 Incentive structures

392 Players receive payoffs based on the underlying incentive structure. We consider five
 393 incentive structures: feedback structure, niche-expert structure, direct intervention
 394 on AI use of these two structures, and a balanced structure.

395 The feedback structure does not directly reward individuals according to their
 396 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 397 residual error of the collective prediction and therefore have the potential to move
 398 the collective prediction toward the true value. Given player A ’s revised prediction
 399 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

400 The niche-expert structure assigns higher payoffs to individuals who make accu-
 401 rate predictions while having an underrepresented interest. Specifically, the squared
 402 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 403 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

404 Previous research has shown that these two incentive structures can produce near
 405 perfect collective accuracy through social learning, except when the niche-expert
 406 incentive is combined with simple averaging (Wang et al., 2025). They are therefore
 407 used as baseline structures for examining how AI assistance affects the emergence of
 408 CI.

409 To examine the effects of directly encouraging or discouraging AI use, we mod-
 410 ify the feedback and niche-expert structures by adding a term proportional to the

411 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

412 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 413 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 414 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 415 the original feedback and niche-expert structures.

416 The balanced structure is defined as the arithmetic mean of the feedback and
 417 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

418 Simulations

419 All simulations are conducted using an agent-based model with 50 causal factors
 420 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 421 target are sampled at the beginning of the simulations and subsequently held fixed.

422 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 423 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 424 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

425 Players’ initial strategies are also sampled from predefined probability distribu-
 426 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 427 simple averaging, each factor’s contribution is scaled by the proportion of players in-
 428 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 429 with mean zero and standard deviation 100, following the scaling used in previous
 430 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 431 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 432 are averaged separately within each interest group and then summed, no additional
 433 scaling of c_A is required. The initial interest of each player is sampled uniformly
 434 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 435 distribution on $[0, 1]$.

436 We assume that prediction events occur on a substantially faster timescale than
 437 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$
 438 of each player analytically rather than explicitly sampling prediction events. The
 439 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 440 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 441 target parameters. Closed form expression of expected payoffs are derived for all
 442 incentive structures and are presented in the Supplementary Methods.

443 Each generation consists of an expected payoff calculation followed by one op-
 444 portunity for pairwise social learning. Two distinct players, A and B , are randomly

sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does not imitate player B but instead redraw all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4.

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy is normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review & Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervision, Writing - Original Draft Preparation, Writing - Review & Editing.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

⁴⁷⁸ **Competing Interests**

⁴⁷⁹ The authors declare no competing interests.

Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

S1 Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

S1.1 Model

Because Omniscient AI observes all M causal factors, its prediction for player A is given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S1})$$

where b_i represent factor-specific biases and ϵ represents stochastic error, following a Gaussian distribution with mean zero and standard deviation τ . A new stochastic error is generated independently for each player and prediction event.

Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S2})$$

In contrast to the AI prediction, player A 's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S3})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S4})$$

represents player A 's interest, c_A represents their belief about the effect of the corresponding factor, and β_A represents their reliance on AI. As in the main text, we set $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

504 S1.2 Aggregation

505 Because the AI component of each revised prediction contains information about all
506 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_A) c_j X_{e_j}\} \quad (S7)$$

507 To separate the contributions of AI and the players' initial predictions, we define the
508 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
509 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

510 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
511 remaining after averaging, and γ_i represents the contribution of players' initial pre-
512 dictions to factor i . Using these definitions, the revised collective prediction can be
513 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

514 Thus, the collective prediction consists of an AI-derived component, the contributions
515 of the players' initial predictions, and the stochastic AI error that remains after
516 aggregation.

517 S1.3 Expected payoffs

518 We next derive the expected payoff under each incentive structure. Expectations are
519 taken over the distributions of the causal factors and stochastic AI errors, conditional
520 on the current population strategies and the fixed parameters of the prediction target.

521 Feedback

522 Under the feedback structure, a player's payoff depends on the alignment between
 523 their revised prediction and the residual error of the collective prediction. To express
 524 this residual, we first define the remaining error in the intercept and in the coefficient
 525 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

526 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 527 remaining error associated with factor i . Because the causal factors and stochastic AI
 528 errors are independent and have zero mean, cross terms vanish when the expectation
 529 is taken. Substituting the revised individual prediction and collective residual into
 530 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^m \delta_i X_i - \bar{\eta} \right\} \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

531 The first term represents the alignment between the AI-derived prediction and the
 532 collective residual, whereas the second term represents the contribution of the player's
 533 initial prediction. The final term arises because the stochastic error in player A 's AI
 534 prediction also contributes to the collective prediction. This finite-population effect
 535 decreases as population size increases.

536 Niche-expert

537 Under the niche-expert structure, the payoff is the negative squared error of the
 538 revised individual prediction, weighted by the frequency of the player's interest.
 539 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

553 S1.4 Expected accuracy

554 Using the residual defined above, the mean squared error of the revised collective
555 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

556 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

557 The collective prediction error consists of the squared error in the intercept, the re-
558 maining error associated with each causal factor, and the stochastic AI error remain-
559 ing after averaging across players. Counterfactual collective accuracy is obtained
560 from the same expression by setting $\beta_A = 0$ for all players while retaining their
561 evolved interests and beliefs.

562 S2 Chatbot AI

563 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each
564 player's interest. This causes the information contained in AI predictions to depend
565 directly on the distribution of interests in the population. Here, we derive the collec-
566 tive prediction, expected payoffs, and expected collective accuracy under this form
567 of AI assistance.

568 S2.1 Model

569 The prediction generated by Chatbot AI for player A depends on the causal factor
570 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

571 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
572 The stochastic error follows a Gaussian distribution with mean zero and standard
573 deviation τ and is generated independently across players and prediction events. The

574 player's initial and revised predictions are given by Player A's initial and revised
 575 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

576 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 577 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 578 tion remains specific to the single factor associated with the player's interest.

579 S2.2 Aggregation

580 Because both the player's initial prediction and the Chatbot AI prediction contain
 581 information about the same causal factor, we use clustering aggregation. Revised
 582 predictions are first averaged among players interested in the same factor and then
 583 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

584 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 585 players in that set.

586 To simplify the collective prediction, we define the effective coefficient, weighted
 587 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$ denotes the set of represented interests.

S2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

608 Niche-expert

609 Under the niche-expert structure, the negative squared error of each revised indi-
 610 vidual prediction is weighted by the frequency of the player's interest. We consider
 611 $e_A = 0$ separately because these players make constant predictions rather than factor-
 612 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

613 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 614 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

615 In both cases, the expected squared error consists of the mismatch between the
 616 player's belief and the true coefficient, contributions from causal factors omitted
 617 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 618 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 619 interest.

620 Incentivizing or penalizing AI use

621 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 622 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

S2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

639 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

640 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
641 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

642 S3 Evolutionary dynamics

643 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,
644 2005), we use replicator dynamics as a deterministic large-population approximation
645 to characterize equilibrium states and their local stability. The agent-based sim-
646 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff
647 stationary states, the replicator and Fermi formulations imply the same local invasion
648 directions for any finite selection strength $s > 0$, although their transient dynamics
649 need not coincide. We focus on the feedback and niche-expert structures, for which
650 the long-term dynamics can be characterized analytically. The direct interventions
651 on AI use and the balanced structure are examined through simulations.

652 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
653 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

654 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
655 internal stationary state, all strategies with positive frequency must have the same
656 expected payoff. A boundary state can also be stationary because a strategy with zero
657 frequency cannot increase unless it is introduced into the population. We therefore
658 assess the stability of boundary states by asking whether they can be invaded by a
659 rare alternative strategy.

660 S3.1 Omniscient AI

661 With Omniscient AI, every AI prediction contains information about the full predic-
662 tion target, and revised predictions are combined through averaging. As population
663 size increases, independent AI errors vanish from the collective prediction. How-
664 ever, these errors remain in individual prediction errors and can therefore continue
665 to affect niche-expert payoffs. We consider the two incentive structures separately.

666 S3.1.1 Feedback

667 Under the feedback structure, both the stochastic error in the collective prediction,
 668 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous
 669 work (Wang et al., 2025), we define the Lyapunov function as the expected error of
 670 collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

671 The function V is non-negative and equals zero when the collective prediction is
 672 identical to the target. To determine how V changes over time, we take its time
 673 derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S78})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S79})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S80})$$

674 For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S81})$$

675 Although K changes with the population state, it is common to all strategies at a
 676 given state and can therefore be treated as a constant with respect to the strategy
 677 index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S82})$$

678 Under the replicator equation, the rate of change in the population mean of a
 679 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S83})$$

680 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S84})$$

681 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S85})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S86})$$

For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S87})$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S88})$$

Because payoff variance is non-negative, the expected collective prediction error cannot increase and decreases whenever the strategies currently represented in the population differ in their expected payoffs. The derivative becomes zero when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

S3.1.2 Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S89})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S91})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S92})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S93})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S94})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S95})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

741 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
742 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S96})$$

743 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
744 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S97})$$

745 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S98})$$

746 This internal stationary state is locally stable. At equilibrium, all represented in-
747 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
748 interest increases slightly above its equilibrium value, its payoff decreases because
749 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
750 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
751 namics therefore generate a restoring force that reduces small deviations in interest
752 frequencies and drives the population back toward the internal stationary state.

753 If one or more interests are absent, the resulting boundary state can be stationary
754 under the replicator equation, but it is unstable against invasion. The payoff of a rare
755 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
756 population is negative when $L_e > 0$, so the rare interest increases when introduced
757 at sufficiently small frequency.

758 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
759 prediction strategies and their frequencies. Whether the constrained optimum has
760 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
761 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
762 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
763 interest frequency remain uniquely determined.

764 Having characterized this equilibrium, we next compare its collective prediction
765 with a prediction made entirely by AI. In particular, the expected squared error of
766 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S99})$$

767 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
768 fore, because L_e is the minimum individual expected squared prediction error within
769 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S100})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S101})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S102})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S103})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S104})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S105})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S106})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S107})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

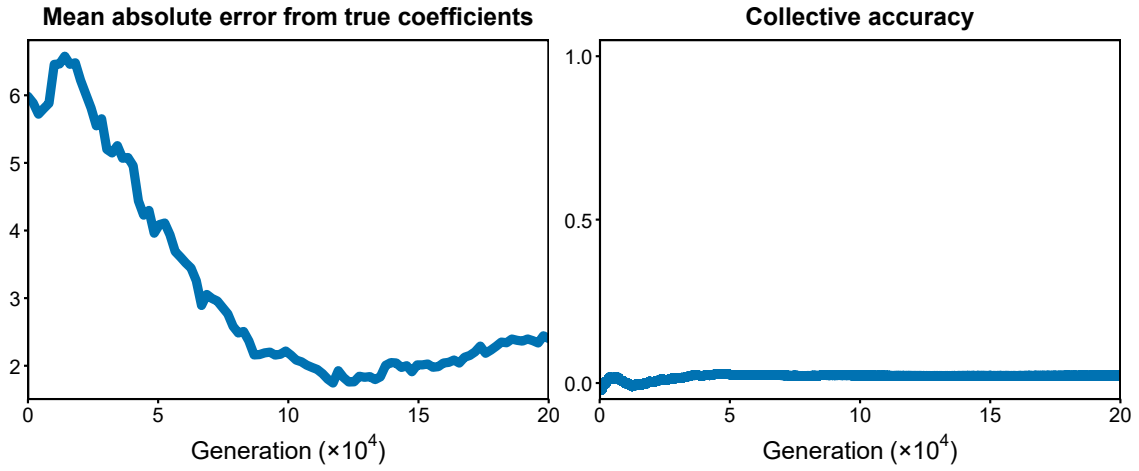


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

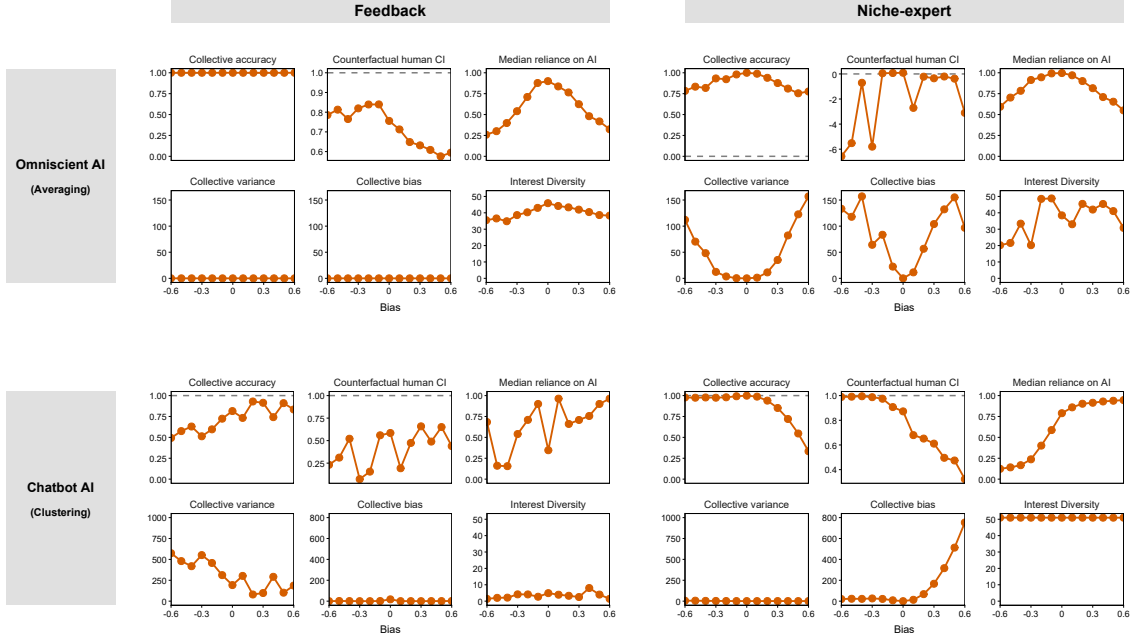


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

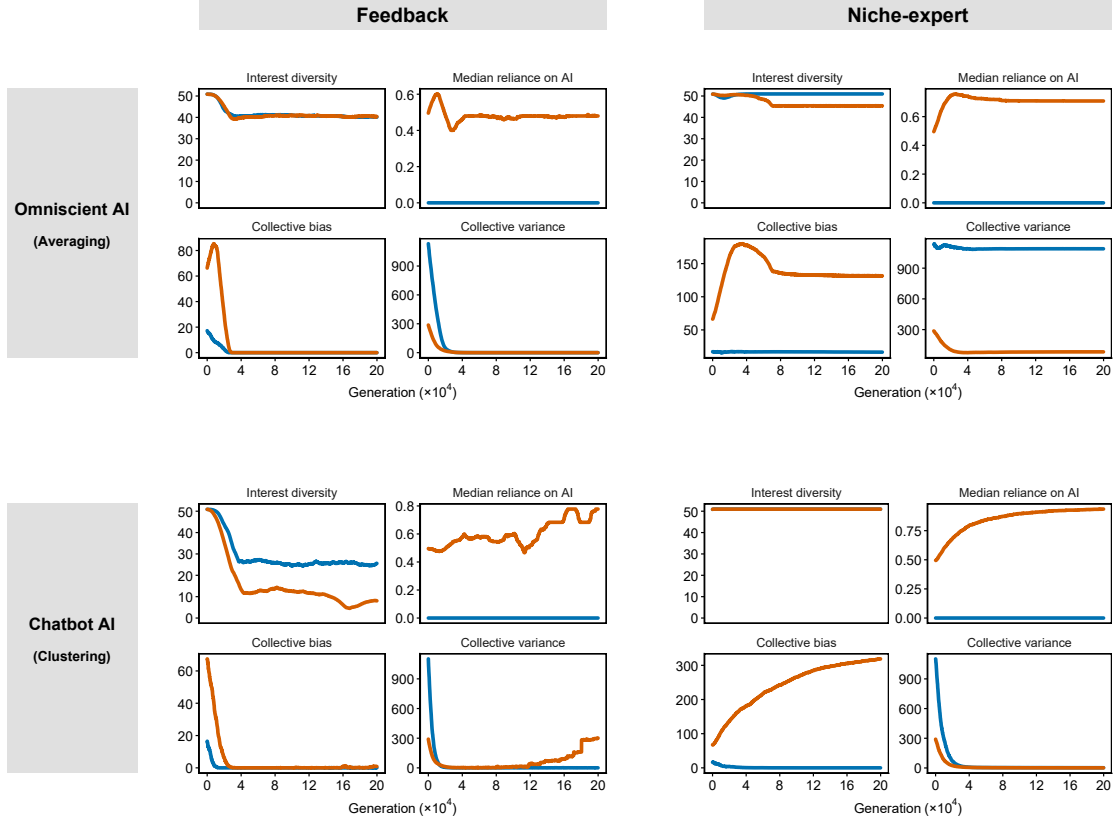


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

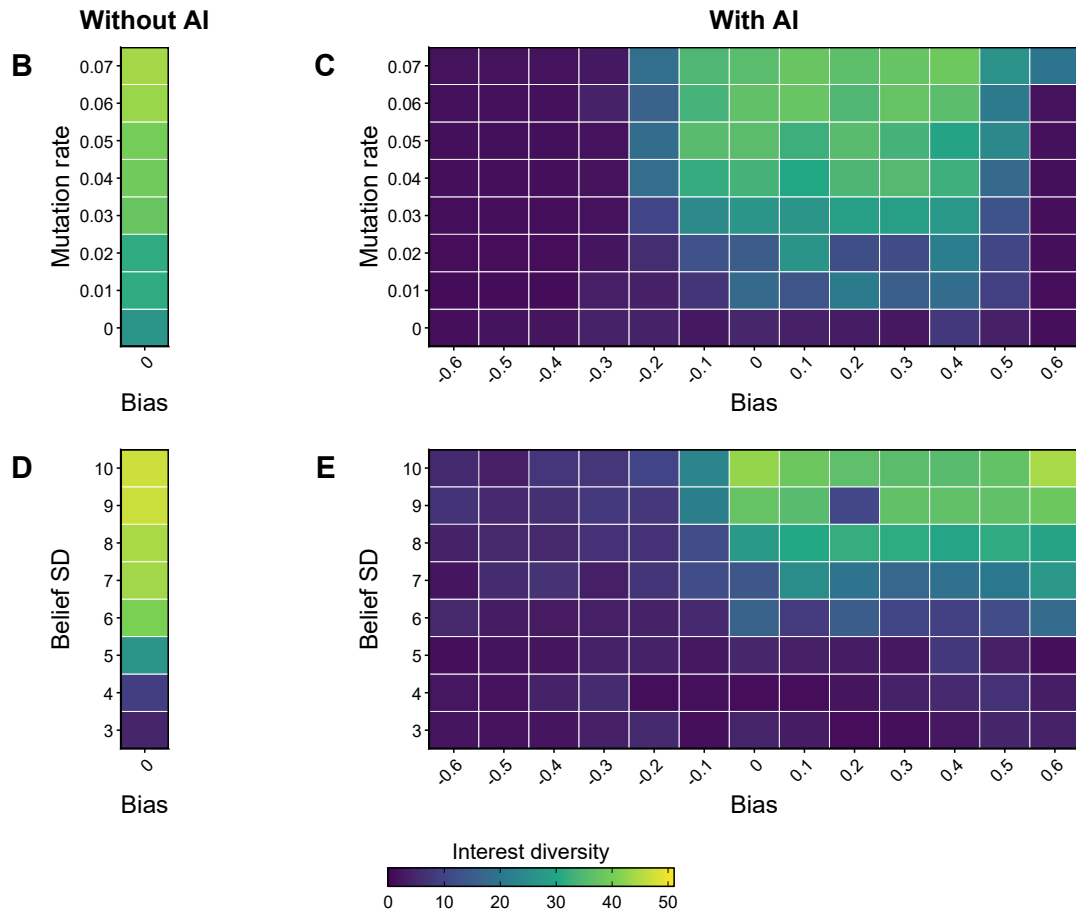
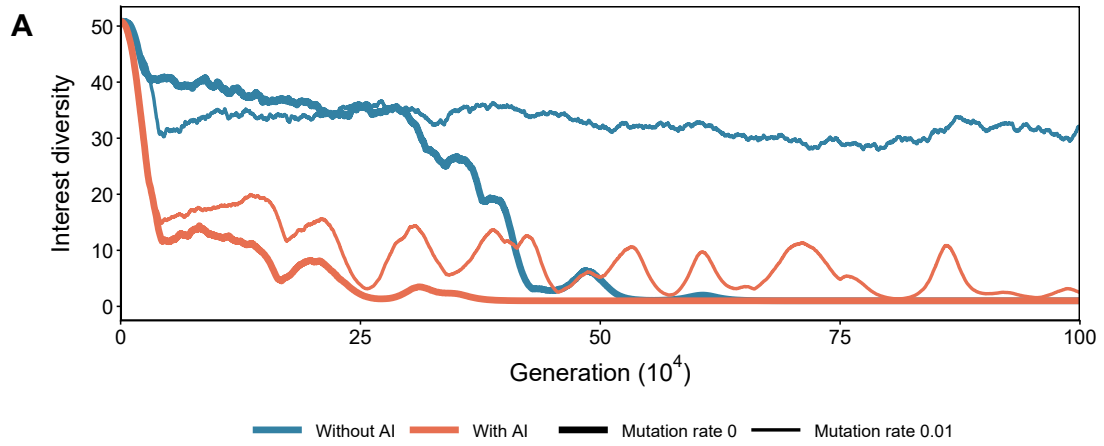


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

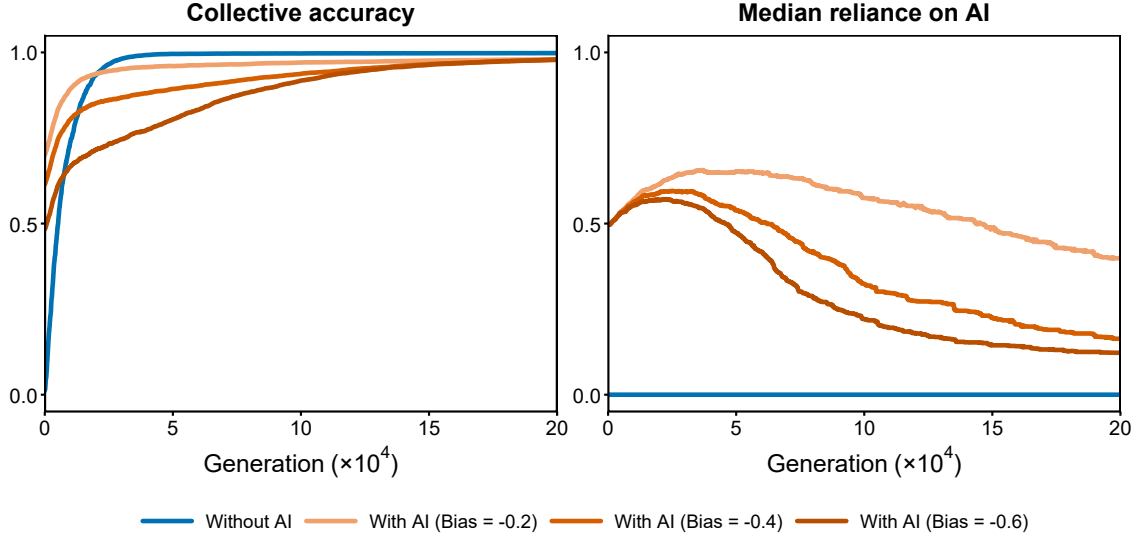


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

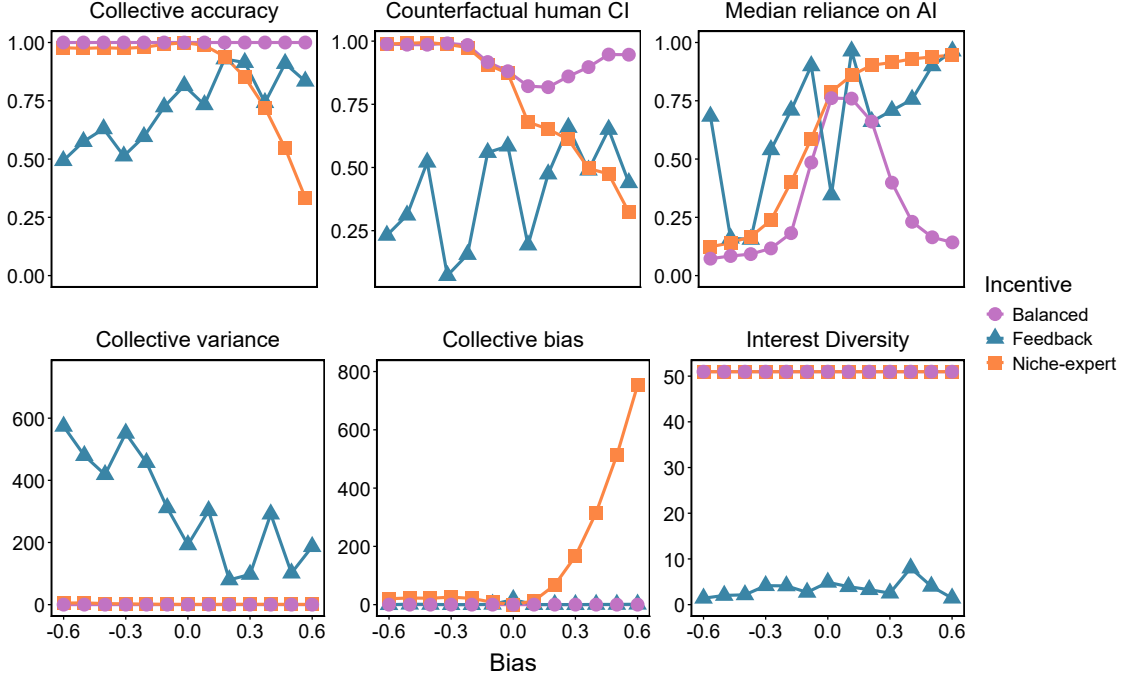


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

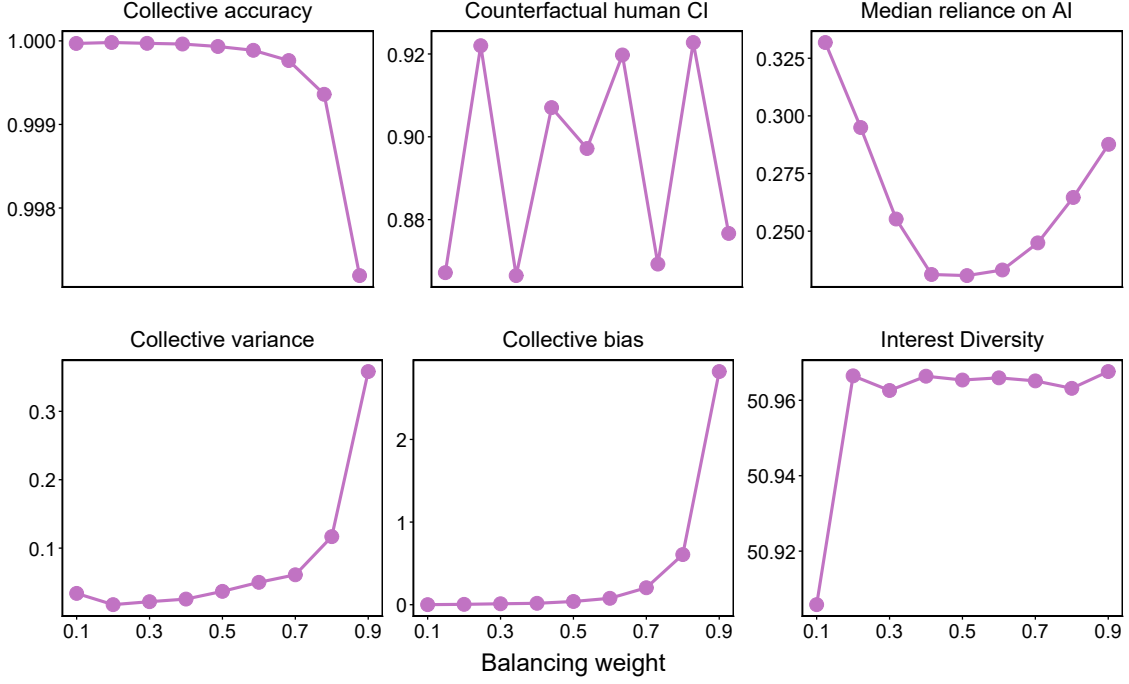


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

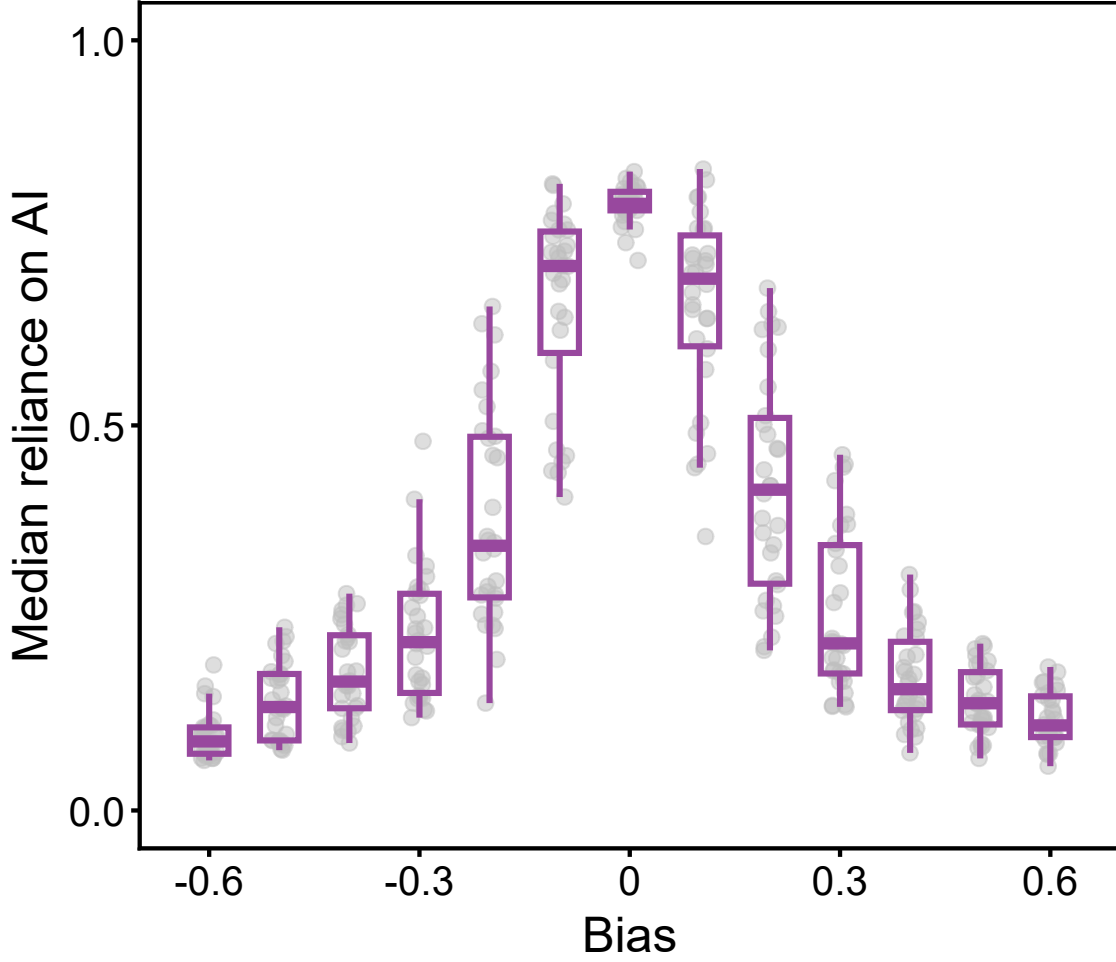


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

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